TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Enabling technologies for all integrated quantum photonic circuits**			
Korea's 12 National Strategic Technology Fields	Quantum (12.48)			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Quantum			
	Medium Category	Small Category	Medium Category	Small Category
	Quantum computing			
Concept of Proposed Technology	We propose to develop **photonic integrated circuits (PICs) for quantum applications** by leveraging our well-established know-how on design and fabrication of Lithium Niobate on Insulator (LNOI) platform. More specifically, we aim at taking advantage of the strong optical non linearities of LNOI for on chip generation of **entangled photon pairs** and for their manipulation via LNOI passive components (interferometers, waveguides, ring resonators…) to achieve quantum computation gates building blocks. In addition, we propose to leverage our expertise in micro-nano fabrication of Silicon Carbide on Insulator (SiCOI) and first demonstration of PIC using this material platform, to achieve **on demand single photon emission and manipulation at room temperature** for applications in quantum computing.			
Rationale for Joint R&D Project	While CSEM is well versed in the design and fabrication of PIC circuits on LNOI and SiCOI platforms, it currently lacks know-how and characterization setups specific to quantum computing applications. In this regard, the participation of a Korean SME active in quantum photonic will greatly complement our current technology offering.			
Key Words	1. Quantum computing 2. Quantum photonics 3. Photonics			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3-4	End of the Project	Level 7-8
R&D Objectives	The objective of the proposal is to demonstrate the key building blocks for quantum photonic computing relying on the specific material platforms LNOI and SiCOI which both offer unique possibilities for a high degree of integration. More specifically: O1: Demonstration of entangled photon pair generation using LNOI to achieve high brightness and visibility. O2: Demonstration of on-demand single photon generation at room temperature using SiCOI			
R&D Target Specifications	More specifically, the proposed quantifiable targets are: T1: Generation of entangled photon pairs in LNOI with visibility > 99%, brightness > 30 MHz nm^{-1} mW^{-1} and coincidence-to-accidental ratio > 100 achieving Bell state fidelity >90% T2: Achievement of electrically controlled, plug-and-play high-brightness SiC single-photon source (SPS) at C-band in SiCOI demonstrating high single-photon purity ($g_{(2)}(0) < 0.08$) and stable spectral emission for optical-photon-qubit applications			
R&D Plan	Both LNOI and SiCOI platforms enable monolithic integration of single and entangled photon sources on a single chip, crucial for secure quantum communication and advanced applications like entanglement-based QKD, teleportation, and quantum repeaters. To support the development of such technologies, we will demonstrate the key building blocks needed for fully integrated photonic platform that can: generate on-chip entangled photon pairs and seamlessly interface SPS color centers and other integrated photonic and electronic circuitry and optical fiber for scalable deployment in quantum communications and networks. Specifically, we will provide the demonstration of: D1) High brightness entangled photon pair generation on chip in the telecom band and D2) quantum PIC chip system integration in a quantum communication testbed, showing entanglement distribution and electrically injected single-photon room temperature generation. For D1, we will design and optimize $\chi(2)$ and $\chi(3)$ nonlinear optical processes in engineered LNOI waveguides and resonators to produce polarization and/or time-bin entangled photon pairs, ensuring compatibility and efficient coupling to the QPIC circuitry developed in the project, pursuing four-wave mixing (SFWM). To achieve this aim, we will build on our extensive QPIC design and experimental capabilities at **CSEM** and by taking advantage of the experience, and know-how on nonlinear quantum photonic integration and characterization at the **Korean SME**. We will also explore new pathways for spectral engineering and nonlinearity enhancement for high brightness and purity of the entangled photon sources in the C-band, afforded by on-chip SBS functionalities. For D2, we will aim at providing a higher-level proof-of-principle demonstrator of the potential of room-temperature SiC QPIC technology, combining the most promising and mature results achieved in the development of the individual building blocks and their co-packaging and integration on a single chip.			

The final demonstrator will be a fully packaged SiCOI Q-PIC with electrical controls to emit single photons. Its performance will be assessed by the **Korean SME** in dedicated experiments. The choice of the actual architecture, level of PIC complexity, functionality integration and co-packaging to be adopted for the PIC chip to be used in the final system demonstrations will be made at the end of month 18, considering the achieved performance, maturity and compatibilities of the different building blocks. In any case, we will envisage by the end of the project the demonstration of quantum gates using entangled photons with high fidelity and the achievement of a deterministic true single photon source.

The overall methodology followed to achieve the above listed goals in the project involves: **Device design & simulation,** including electromagnetic design of waveguides, micro-rings, photonic crystal cavities for relevant operation wavelengths; optimized dispersion, birefringence and mode engineering for efficient nonlinear interactions considering both $\chi(2)$ and $\chi(3)$ capabilities both LNOI and SiCOI as well as the spectral features of relevant SPS defects; **Nanofabrication & basic nonlinear characterization** of low-loss QPIC building blocks of the relevant nonlinear conversion modules and matching circuitry for tuning, enhancement and filtering, to assess efficiency, noise and spectral performances. **Quantum measurements**, involving room-temperature confocal spectroscopy for emitter identification (PL spectrum, brightness) and on chip $g^2(0)$ measurements to confirm single-photon purity; Hong-Ou-Mandel and quantum state tomography to test indistinguishability and entanglement for generated pairs, measurement of internal/external conversion efficiency and pair generation rate, added background, Coincidence-to-Accidental Ratio (CAR), joint spectral density.

WP1: Generation of entangled photons using LNOI (Month 1- Month 24)

The aim of this WP is the design, fabrication, and characterization of non-classical light entangled photon sources (EPS) LNOI integrated circuits.

T1.1: Entangled and heralded photon sources for the telecom (Month 1- M18): with the support of the **Korean SME**, **CSEM** will simulate, and design engineered LNOI photonic waveguides and ring resonator to achieve spontaneous four wave mixing (SFWM) for correlated photon-pair emission in the 1.55 µm (telecom) band. We will target on-chip photon-pair source brightness > 30 MHz nm^{-1} mW^{-1} achieving two photon interference visibility > 99%, coincidence-to-accidental ratio > 100 and achieving Bell state fidelity >90%. To achieve such results, we will optimize both design and micro-fabrication processes following feedback from individual building block characterization at **CSEM** and quantum properties characterization in T1.2

T1.2: Entangled photon source characterization (M12-M24): The **Korean SME** will also design suitable quantum characterization setup and use it to qualify the sources developed in T1.1. In addition, the SME will propose a layout for quantum state manipulation which will be also fabricated in T1.1.

Following interactions between T1.1 and T1.2, and if further required for the targeted application, we will explore the addressability of the quantum source in the C-band with ~10 nm tunability, and purification by Stimulated Brillouin Scattering-based filtering to below 1 nm.

WP2: On demand generation of single photon in SiCOI (Month 1- Month 18)

WP2 primary goal is to engineer and implement color centers and related quantum emitters within SiC, ensuring compatibility with room-temperature operation and scalable photonic integration. This WP will address: Color-Center creation and electrical control, SBS-based filter design and characterization to have a narrow linewidth, pure filter, Integration of SPS and PIC for color correction and to have a coherent SPS, room temperature operation:

T2.1: Coherent Single Photon Source (SPS)-based Q-PIC design and fabrication (M1-M18): In this task, **CSEM** will study and identify the processes for deterministic implantation or in-situ growth of color centers (e.g., divacancies) in SiC with high spatial precision and controlled density, to emit a single photon at room temperature. On its side, the **Korean SME** will characterize their emission properties (linewidth, lifetime, coherence) to provide feedback to CSEM in order to optimize design and fabrication conditions. In addition, CSEM will assess the implementation of a P-I-N junction for electrically driving SPS by doping the SiC region hosting the color center, evaluating both pre- and post-color-center integration approaches addressing the implanted color centers to open the future perspectives of achieving fully electrically controlled single photon emission on the SiCOI platform.

T2.2: SPS characterization (M12-M24): In this task, the **Korean SME** will characterize the best packaged SiCOI SPS to provide detailed feedback on performance under real quantum conditions, ensuring iterative improvements to individual devices and overall system functionality. The Korean SME will notably measure the purity and indistinguishability of the emitted photon via Hanbury-Brown-Twiss interferometry and Hong–Ou–Mandel-type two-photon interference, respectively. While indistinguishable photon generation in SiCOI has not yet been demonstrated, we aim to demonstrate such a property, possibly by coupling the SPS with active frequency filtering devices. We also aim to show high photon purity emission at room temperature with a $g_{(2)}(0)<0.08$.

The project will have the following deliverables:

D1: Packaged LNOI quantum PIC chip generating entangled photon pairs and containing the optical circuit for quantum state manipulation. The chip will be ready for optical I/O connectivity following the layout required by the Korean SME. (M24)

D2: SiCOI chip containing electrically driven single photon emitting source with on chip coupling to photon manipulation blocks which will be agreed following discussion with the SME. (M24)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	It is expected that CSEM will carry out all the design and micro fabrication activities on both LNOI and SiCOI material platforms. In addition, CSEM will perform “standard” optical characterization of the PICs such as propagation and insertion loss measurements, modulation efficiency, harmonic generation characterization, etc..
Expected R&R of Korean SME Partner	It is expected that the Korean SME will assist CSEM in design of quantum PIC and carry out quantum specific measurements such as single photon purity and indistinguishability measurements as well as entangled photon source brightness and coincidence measurements.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	It is expected that the project will bring fundamental elements for the achievement of all integrated quantum photonic circuits which will find applications not only in quantum computing but also in quantum communication. Of specific interest is the development of a potential platform operating at room temperature and relying on a monolithically integrated development with a single material for quantum state generation and computation. This will represent significant advantages on the existing industry state-of-the-art often relying on cryogenic-based systems (e.g. with superconducting qubits) or on heterogeneous integration and off-the-chip photon generation leading to more expensive fabrication processes as well as detrimental coupling losses. Eventually, owing to the good radiation hardness and thermal dissipation of SiC, the possibility to achieve both quantum PIC and sensing with SiCOI also opens unique prospects for their integration in harsh environment applications such as space. It is expected that following these developments, the Korean SME will possess a strong edge over its existing international competitors and will be ideally placed to leverage the acquired know-how to extend its activities in various fields of quantum technologies. The second quantum revolution is driving transformative technologies in computing, communication, and sensing, with global markets projected to exceed €30B by 2040 for quantum computing alone. However, most current quantum systems rely on cryogenic cooling, which imposes severe limitations on cost, complexity, and scalability. For quantum technologies to become practical and widely deployed, a room-temperature, CMOS-compatible quantum photonic platform is essential. Our proposal addresses this need by introducing LNOI and SiCOI-based QPICs that integrate photon generation, manipulation, and transformation on a single chip; all components except the detector will then be able to work at room temperature. This breakthrough will enable secure quantum communication, high-precision sensing, and scalable quantum computing, while drastically reducing energy consumption and operational costs. Beyond computing, the impact extends to sectors such as telecommunications, healthcare, and space, where robust, miniaturized quantum systems are critical.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization